PRINT JOB

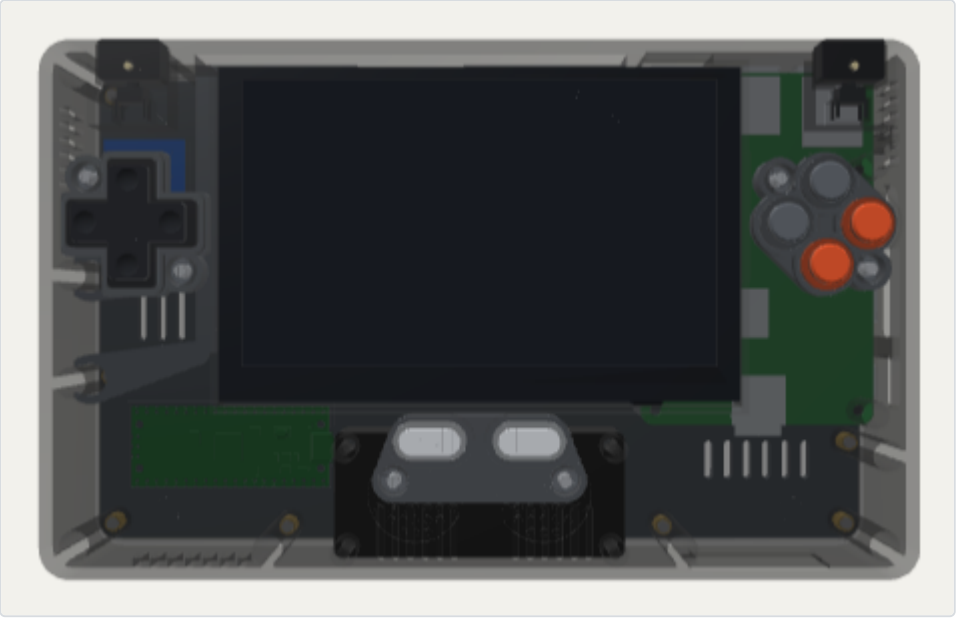
ARCA handheld

Printing guide for the 3D printed enclosure

VERSION	STATUS	DATE
3.1	Ready to print	25 Sep 2026

A handheld games console: a smoky see-through front shell, a hidden inner frame, buttons in four colours and four swappable two-colour back panels.

Item	Detail
Files to print	15, numbered 01 to 15 in the order to print them. Files 16 to 19 replace the panel files on a Bambu Lab P2S with AMS
Largest part	202 x 124 x 44 mm. Needs a bed of at least **205 x 130 mm**
Filament	Up to about 750 g if printed solid. Expect 500 to 600 g with the settings below
Supports	**None.** Every part is designed to print without them
Orientation	**Already set.** Load each file as it is. Do not rotate, flip or scale



Before you start

1. **Print file 01, the fit-check jig, first** and hand it to the owner. It carries every hole and post position so the real parts can be checked on it. **Wait for the go-ahead before printing file 02, the front shell**, which is the long print.
2. **Supports OFF.** Automatic supports would fill the switch pockets, screw holes and slots.
3. **Do not scale any file** to fix hole sizes. If holes come out small, use the slicer's XY hole compensation instead (see the quality check).

Slicer settings

These apply to every part unless the parts list says otherwise.

Setting	Value	Why
Nozzle	0.4 mm	
Layer height	0.20 mm	Buttons: 0.12 to 0.16 mm if you can, for a smoother face
Walls / perimeters	**4** (1.6 mm)	The screw bosses and posts need solid walls
Top and bottom layers	5	
Infill	25 % gyroid	Switch plates and the D-pad set: 40 %
Supports	**Off**	
Brim	5 mm	Front shell, rear frame and the four panels (large flat parts)
Seam	Aligned or rear	

Material

- **PETG is preferred for the body parts** (files 02 to 07). The Raspberry Pi inside runs warm, and PLA softens from about 55 C.
- **PLA is fine for the buttons and the four panels** (files 08 to 19).
- If PLA is all you have, everything will work for a demo. Just keep the device out of hot cars and direct sun.

Parts list

The number at the front of each file name is its print order.

#	File	Pieces	Colour	Notes
01	`fit_check_jig`	1	Any	**Print first.** Fast settings fine, but keep 4 walls
02	`front_shell`	1	**Smoke** (translucent black)	The visible front. Face down as exported. Brim on. Longest print
03	`sd_hatch`	1	**Smoke**	Tiny. Print it alongside another part
04	`rear_frame`	1	Any (hidden)	Brim on
05	`bay_tray`	1	Any (hidden)	
06	`switch_plates`	3	Any (hidden)	Infill 40 %
07	`display_spacers`	4	Any (hidden)	
08	`buttons_A_B`	2	**Orange** (#E4572E)	Caps print face down: leave as is
09	`buttons_X_Y`	2	**Grey** (#5F646B)	
10	`buttons_START_SELECT`	2	**Light grey** (#C9CCD1)	
11	`dpad_and_shoulders`	3	**Black / charcoal** (#26282C)	Infill 40 %
12	`panel_world`	1	Two colours, see below	
13	`panel_arkiv`	1	Two colours, see below	
14	`panel_arc`	1	Two colours, see below	
15	`panel_chainlink`	1	Two colours, see below	
16 to 19	`AMS_Bambu_P2S_panel_...` (.3mf)	1 each	Colours set in the file	Bambu Lab P2S with AMS only, instead of 12 to 15. See below

#	File	Pieces	Colour	Notes
20	`panel_ethglobal_mumbai`	1	Black, orange logo	Single colour logo; change filament at Z = 4.0 mm
21	`AMS_Bambu_P2S_panel_ethglobal_mumbai` (.3mf)	1	Black + orange, red, green	Four colours: needs all four AMS slots
22	`panel_ethglobal_mumbai_lockup`	1	White, red logo	Diya + "ETHGlobal MUMBAI"; single colour logo, change filament at Z = 4.0 mm
23	`AMS_Bambu_P2S_panel_ethglobal_mumbai_lockup` (.3mf)	1	White + orange, red, green	The full lockup, 4 slots
24	`panel_hedera`	1	Two colours, see below	Change filament at Z = 4.0 mm
25	`AMS_Bambu_P2S_panel_hedera` (.3mf)	1	Colours set in the file	Bambu Lab P2S with AMS only, instead of 24

If no translucent black is available, solid black is an acceptable fallback for files 02 and 03.

The two-colour panels

Each panel is a 4.0 mm plate with a logo raised 0.8 mm on top. The flat side, with small magnet holes and a groove, goes on the bed, exactly as exported.

Panel	Plate colour	Logo colour
World (12)	White	Black
Arkiv (13)	Blue	White
Arc (14)	Black	White
Chainlink (15)	Blue	White
Hedera (24, 25)	Black	White
ETHGlobal Mumbai (20, 21)	Black	Orange; on file 21 also red and green
ETHGlobal Mumbai lockup (22, 23)	White	Red; on file 23 also orange and green

Single nozzle. Print files 12 to 15 and 24 and add a filament change at **Z = 4.0 mm**. Use pause, M600 or "colour change", whichever your printer has. That is the first layer of the logo, the one printing from 4.0 to 4.2 mm, which is layer 21 with 0.2 mm layers.

Bambu Lab P2S with AMS. Use files 16 to 19, 21, 23 and 25 instead of the single-colour files. Each is a ready Bambu Studio project: the panel is one object whose logo part is fused to its face, already sliced and set up for:

Setting	Value
Printer	Bambu Lab P2S, 0.4 mm hardened steel nozzle
Plate	Textured PEI Plate
Process	0.20mm Standard @BBL P2S, 4 walls, 5 top and 5 bottom layers, 25 % gyroid, 5 mm outer brim, no supports, prime tower off
Filament	Bambu PLA Basic. Slot 1 = plate colour, slot 2 = logo colour, as in the table above
Per panel	About 2 h, 80 g plate colour + 1 to 2.5 g logo colour, one filament change (at 4.0 mm)

Open the file in Bambu Studio, click Print plate, and map the two filaments to the matching AMS slots. There is only one filament change, so no prime tower: the purge goes to the chute.

Other multi-material printers. Open the .3mf in your slicer (Orca, Prusa and others read Bambu projects) and use the object's body and logo parts, or ask for split STLs.

Quality check before handing over

If you have calipers, these are the dimensions that matter:

Where	Should measure
Front shell, outside	202 x 124 mm
The 8 tall round posts inside the shell: hole	4.1 mm (heat-set inserts go in here)
Rear frame: 8 screw holes / their counterbores	3.3 mm / 6.4 mm
Rear frame, flat outer side: 6 magnet pockets	6.3 mm wide, 3.15 mm deep
Switch plates: square pockets	6.3 mm
Shell, the two blocks inside the top edge: small vertical hole	1.95 mm (a 1.75 mm filament must drop through freely)
Shell, the two slots in the top edge	15.4 mm wide

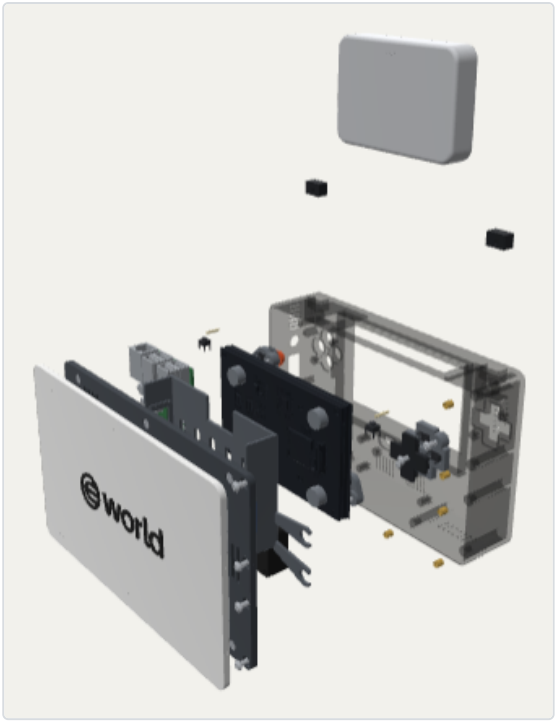
- **Holes printing small** is normal with PETG. Set XY hole compensation to +0.1 mm and reprint that part. Never scale the model.
- **The two clips.** On the inside of the shell wall that has the HDMI hole, there are two thin upright clips with a small lip at the top. Check they are intact and not strung together with the wall.

Please also include

- **About 5 cm of 1.75 mm filament**, any colour. It gets cut into two 10.5 mm pins that hold the shoulder buttons in.

How the parts fit together

For reference only. The owner does the assembly.



`view_all_parts_in_3d.html` opens in any web browser. It shows every part in 3D, the assembled device, and a step-by-step assembly.